

NeuroFlow CFD Analyst

Clinical Hemodynamic Assessment

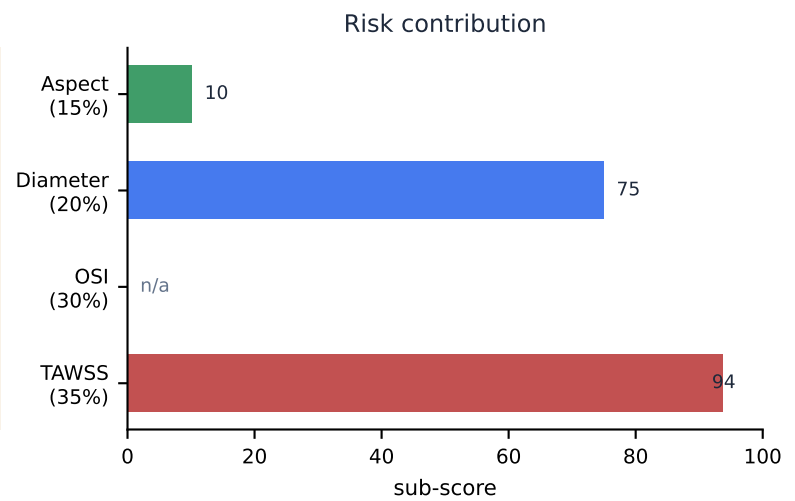
Case PT-2026-0101 · generated 2026-08-05 08:15

49

Composite Risk Index / 100

Moderate

Lower bound — OSI (30%) not computed on this steady solve



Hemodynamic parameters

Parameter	Value	Threshold	Status
TAWSS — aneurysm dome	0.235 Pa	< 0.4 Pa	FLAGGED
TAWSS — parent artery	2.962 Pa	reference	normal
OSI — aneurysm dome	not computed	steady solve	—
RRT — aneurysm dome	11.07 Pa ⁻¹	> 3.0	FLAGGED
ECAP — aneurysm dome	not computed	steady solve	—
NWSS (sac / parent)	0.080	—	normal
LSAR (relative, <10% parent)	84.3 %	—	normal
LSAR (absolute, <0.4 Pa)	88.2 %	—	normal

Morphology (measured from the reconstructed surface)

Max diameter	Neck diameter	Aspect ratio	Dome-to-neck
8.00 mm	6.90 mm	0.88	1.16
Volume	Surface area	Non-sphericity	
240.7 mm ³	174.1 mm ²	0.147	

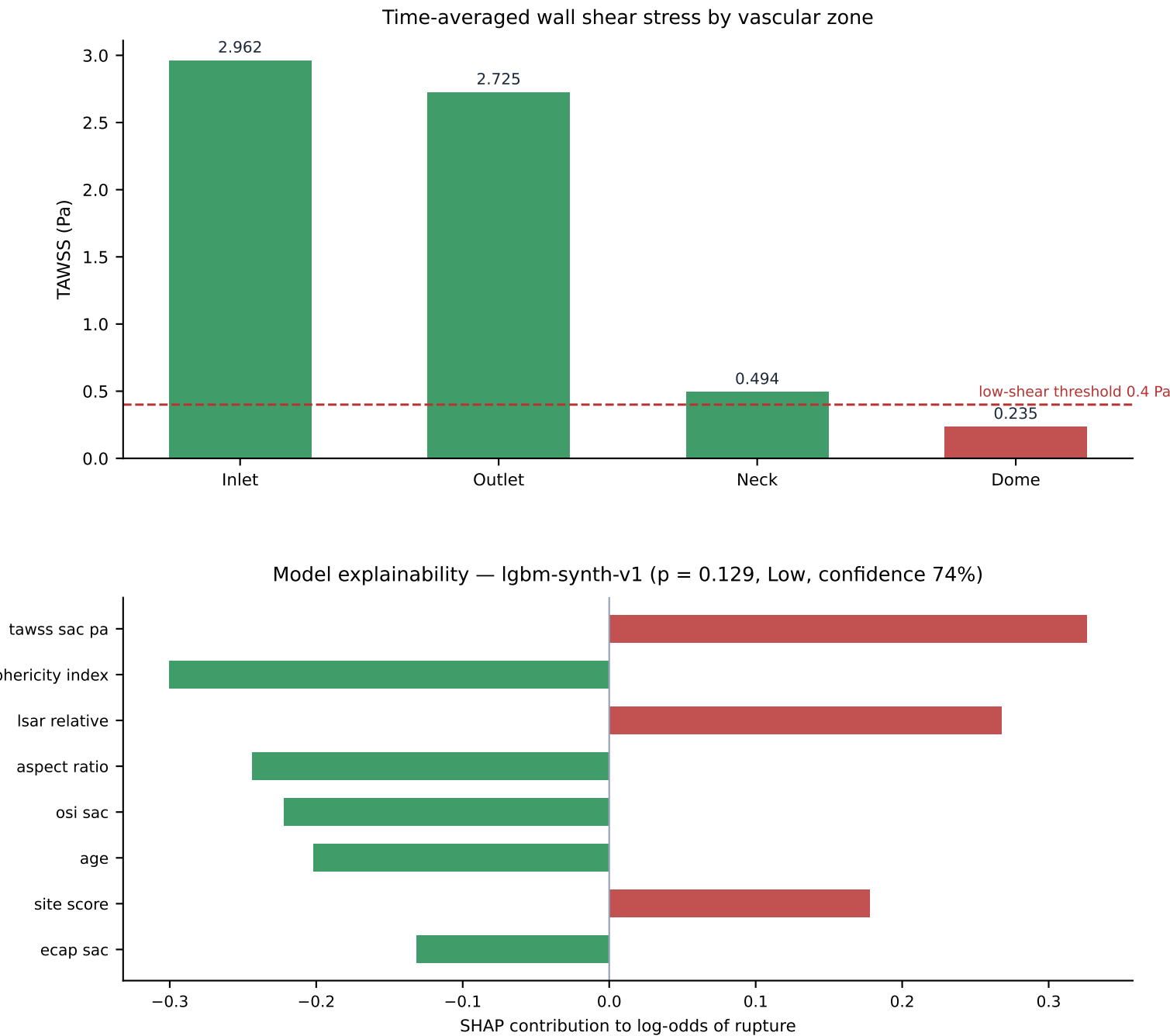
Provenance

Solver: OpenFOAM ESI v2412 · Mesh: cells: 511015 ·
WSS converted from OpenFOAM kinematic units (m²/s²) to Pascals by multiplying by rho = 1060.0. Validated against the analytic Poiseuille solution $\tau = 4 \cdot \mu \cdot Q / (\pi \cdot r^3)$.

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Zone Analysis and Model Explainability

Case PT-2026-0101



Assessment

Incomplete input vector: OSI and ECAP were not computed for this steady solve and entered the model as zero.

Computational fluid dynamics analysis of case PT-2026-0101. Time-averaged wall shear stress at the aneurysm dome measured 0.235 Pa against 2.962 Pa in the parent artery (normalised WSS 0.079), with an oscillatory shear index of 0.000. 84.3% of the sac surface lies below 10% of the parent-artery shear (LSAR). Relative residence time at the dome is 11.07 Pa⁻¹. Sac morphology measured from the reconstructed surface: maximum diameter 8.00 mm, neck 6.90 mm, aspect ratio 0.88, volume 240.7 mm³. Composite Risk Index 49/100 (Moderate).